

Clifford algebra

In this note we introduce the rudiments of Clifford algebra. For more details we refer to [LM89, Chapter I].

1. Basic definitions

1.1. Let $\mathbf{k}$ be a field (with characteristic different from 2) and let V be a $\mathbf{k}$ -vector space. A quadratic form is a symmetric, bilinear map $\langle -, - \rangle: V \times V \rightarrow \mathbf{k}$ which is non-degenerate in the sense that $\langle v, w \rangle = 0$ for all $w \in V$ implies $v = 0$. We will write $q(v) = \langle v, v \rangle$ in what follows.

The **Clifford algebra** $\mathcal{Cl}(V, q)$ associated to V is the quotient of the tensor algebra

$$(1) \quad T(V) = \bigoplus_{k \geq 0} V^{\otimes k}$$

by the two-sided ideal generated by elements

$$(2) \quad v \otimes v - q(v) \cdot 1, \quad v \in V.$$

Notice that the relation $v^2 = -q(v)1$, for all $v \in V$, can be equivalently written as the relation

$$(3) \quad v \cdot w + w \cdot v = -2\langle v, w \rangle 1$$

for all $v, w \in V$.

Proposition 1.1. *The Clifford algebra $\mathcal{Cl}(V, q)$ is the universal algebra for which:*

- *there is an injection $i: V \hookrightarrow \mathcal{Cl}(V, q)$,*
- *let $\phi: V \rightarrow A$ be a linear map of V into a (unital) $\mathbf{k}$ -algebra A such that*

$$\phi(x)^2 = -q(x)1.$$

Then there exists a unique homomorphism $\tilde{\phi}: \mathcal{Cl}(V, q) \rightarrow A$ such that $\tilde{\phi} \circ i = \phi$.

The orthogonal group $O(V, q)$ are the linear automorphisms of V which preserve q . If $f: V \rightarrow V$ is such an automorphism then, $f(v)^2 = -q(f(v))1 = q(v)1$ in $\mathcal{Cl}(V, q)$

for all $v \in V$. Thus f defines a unique algebra homomorphism $\tilde{f}: \mathcal{Cl}(V, q) \rightarrow \mathcal{Cl}(V, q)$ with the property that $\tilde{f}|_V = f$. Moreover, since f is bijective so is $\tilde{f}$. Thus we have constructed a group embedding

$$(4) \quad O(V, q) \hookrightarrow \text{Aut } \mathcal{Cl}(V, q).$$

In fact, the group lies within the subgroup of *inner* automorphisms.

1.2. A *filtered* algebra is an algebra A with an exhaustive sequence of subspaces

$$0 = F^{-1}A \subset F^0A \subset \cdots \subset F^\ell A \subset \cdots \subset F^\infty A = A$$

such that if $a \in F^p A, b \in F^q A$ then $ab \in F^{p+q} A$.

Let $A = \{F^\bullet A\}$ be a filtered algebra. The associated graded $\text{gr } A$ has underlying vector space

$$\text{gr } A = \bigoplus_{k \geq 0} F^k A / F^{k-1} A.$$

The product on A defines a product on $\text{gr } A$, giving $\text{gr } A$ the structure of an algebra for which the canonical map $A \rightarrow \text{gr } A$ is a homomorphism.

There is a filtration on the tensor algebra $T(V)$ defined by

$$(5) \quad F^p T(V) \stackrel{\text{def}}{=} \bigoplus_{k \leq p} F^k V.$$

This induces a filtration on the Clifford algebra $\mathcal{Cl}(V, q)$ such that

$$(6) \quad \text{gr } \mathcal{Cl}(V, q) \cong \wedge V,$$

where $\wedge V$ is the exterior algebra on V . This implies the following.

Lemma 1.2. Suppose $\{e_i\}$ is a basis for V . Then

$$e_{i_1} \cdots e_{i_k},$$

where $i_1 < \cdots < i_k, k \geq 0$, form a basis for $\mathcal{Cl}(V, q)$. In particular, $\dim \mathcal{Cl}(V, q) = 2^{\dim V}$.

1.3. Let $\mathcal{Cl}^{ev, odd}(V, q)$ be the images of

$$\bigoplus_{i \geq 0} V^{\otimes 2i}, \quad \bigoplus_{i \geq 0} V^{\otimes 2i+1}$$

in $\mathcal{Cl}(V, q)$, respectively.

Proposition 1.3. Both $\mathcal{Cl}^{ev, odd}(V, q)$ are subalgebras of $\mathcal{Cl}(V, q)$ and

$$(7) \quad \mathcal{Cl}(V, q) = \mathcal{Cl}^{ev}(V, q) \oplus \mathcal{Cl}^{odd}(V, q).$$

Furthermore, the product decomposes as

$$\begin{aligned}\mathcal{C}\ell^{ev}(V, q) \times \mathcal{C}\ell^{ev}(V, q) &\rightarrow \mathcal{C}\ell^{ev}(V, q) \\ \mathcal{C}\ell^{ev}(V, q) \times \mathcal{C}\ell^{odd}(V, q) &\rightarrow \mathcal{C}\ell^{odd}(V, q) \\ \mathcal{C}\ell^{odd}(V, q) \times \mathcal{C}\ell^{ev}(V, q) &\rightarrow \mathcal{C}\ell^{odd}(V, q) \\ \mathcal{C}\ell^{odd}(V, q) \times \mathcal{C}\ell^{odd}(V, q) &\rightarrow \mathcal{C}\ell^{ev}(V, q).\end{aligned}$$

The axioms of the above proposition characterize what is called a $\mathbf{Z}/2$ graded algebra, or *superalgebra*. Notice that $\wedge V$ is naturally a $\mathbf{Z}/2$ graded algebra, and the canonical homomorphism $\mathcal{C}\ell(V, q) \rightarrow \wedge V$ preserves the $\mathbf{Z}/2$ gradings.

Proposition 1.4. *There is a natural isomorphism*

$$(8) \quad \mathcal{C}\ell(V_1 \oplus V_2, q_1 \oplus q_2) \cong \mathcal{C}\ell(V_1, q_1) \otimes^{gr} \mathcal{C}\ell(V_2, q_2)$$

where the tensor product is the **graded** tensor product (see below) of $\mathbf{Z}/2$ graded algebras.

The graded tensor product $A \otimes^{gr} B$ of $\mathbf{Z}/2$ graded algebras A, B differs from the usual tensor product of plain ungraded algebras. As a vector space, it does agree with the standard tensor product

$$A \otimes^{gr} B = A^{ev} \otimes B^{ev} \oplus A^{ev} \otimes B^{odd} \oplus A^{odd} \otimes B^{ev} \oplus A^{odd} \otimes B^{odd}.$$

The product, on the other hand, is defined by

$$(a \otimes x) \cdot (y \otimes b) = (-1)^{|x||y|} ay \otimes xb$$

where $a, y \in A$ and $b, x \in B$.

1.4. Here is an alternative description of the $\mathbf{Z}/2$ grading. Let $i_q: V \hookrightarrow \mathcal{C}\ell(V, q)$ denote the canonical morphism. Consider the automorphism

$$(9) \quad \alpha: \mathcal{C}\ell(V, q) \rightarrow \mathcal{C}\ell(V, q)$$

which extends the linear map $v \mapsto -v$. Since $\alpha^2 = \mathbb{1}_{\mathcal{C}\ell(V, q)}$ we have a decomposition

$$(10) \quad \mathcal{C}\ell^{ev, odd}(V, q) = \{x \in \mathcal{C}\ell(V, q) \mid \alpha(x) = (-1)^{ev, odd} x\}.$$

These are exactly the even/odd subspaces from above.

1.5. As another example of an involution consider the reversal of order map

$$(11) \quad v_1 \otimes \cdots \otimes v_k \mapsto v_k \otimes \cdots \otimes v_1.$$

This preserves the defining ideal so descends to a linear automorphism of the Clifford algebra. This automorphism is not compatible with the algebra structure in the usual sense. It is an anti-automorphism in the sense that $(\varphi\psi)^t = \psi^t\varphi^t$.

2. Pin and spin groups

2.1. Given any algebra A we let $A^\times \subset A$ denote the group of units; the group of elements which admit a multiplicative inverse. There is a group homomorphism

$$(12) \quad \text{Ad}: A^\times \hookrightarrow \text{Aut } A,$$

defined by $\text{Ad}_a: x \mapsto axa^{-1}$.

In the case of the Clifford algebra $A = \text{Cl}(V, q)$ (and $\mathbf{k} = \mathbf{R}$ or $\mathbf{C}$) the group of units $\text{Cl}(V, q)^\times$ is a Lie group of dimension 2^n . The following is a useful computation.

Proposition 2.1. *Suppose $v \in V$ satisfies $q(v) \neq 0$. Then*

$$(13) \quad -\text{Ad}_v(x) = x - 2 \frac{\langle v, x \rangle}{\langle v, v \rangle} v.$$

The Lie algebra $\text{Lie Cl}(V, q)$ is isomorphic to $\text{Cl}(V, q)$ as a vector space and the bracket is the commutator

$$(14) \quad [x, y] \stackrel{\text{def}}{=} xy - yx.$$

(In fact, any algebra A defines a Lie algebra by the commutator.) The derivative of the group-level adjoint defines a Lie algebra homomorphism

$$(15) \quad \text{ad}: \text{Lie Cl}(V, q) \rightarrow \text{Der Cl}(V, q),$$

given by $\text{ad}_y(x) = [y, x]$.

2.2. The orthogonal group $O(V, q) \subset GL(V)$ is the subgroup of linear isomorphisms $A: V \rightarrow V$ which preserve the bilinear form $q(Av) = q(v)$. An easy calculation implies that if $A \in O(V, q)$ then $\det A = \pm 1$. The subgroup $SO(V, q) \subset O(V, q)$ consists of elements with $\det A = 1$. This subgroup is connected.

The Lie algebra of $SO(V, q)$ is the Lie algebra of skew-symmetric matrices

$$(16) \quad \mathfrak{so}(V) = \{A: V \rightarrow V \mid \langle Av, w \rangle = -\langle v, Aw \rangle\}.$$

Proposition 2.2. *The map*

$$(17) \quad T: \wedge^2 V \rightarrow \mathfrak{so}(V)$$

which sends $x \wedge y \in \wedge^2 V$ to the endomorphism

$$(18) \quad T_{x \wedge y}(v) = \langle x, v \rangle y - \langle y, v \rangle x$$

is an isomorphism.

Explicitly, matrix commutator corresponds to the operation on $\wedge^2 V$:

$$(19) \quad [u \wedge v, x \wedge y] = \langle u, x \rangle v \wedge y - \langle u, y \rangle v \wedge x - \langle v, x \rangle u \wedge y + \langle v, y \rangle u \wedge x.$$

Thus, with this bracket, we can identify $\wedge^2 V \cong \mathfrak{so}(V)$ as Lie algebras. Notice that we can write

$$(20) \quad [u \wedge v, x \wedge y] = T_{u \wedge v}(x) \wedge y - T_{u \wedge v}(y) \wedge x.$$

Proposition 2.3. *The Lie algebra $\mathfrak{so}(V)$ naturally embeds into the Clifford algebra via the homomorphism*

$$(21) \quad \rho: \wedge^2 V \cong \mathfrak{so}(V) \rightarrow \mathcal{Cl}(V, q)$$

defined by

$$(22) \quad \rho(u \wedge v) = \frac{1}{4}(uv - vu).$$

To see that this is a homomorphism we need to see that

$$(23) \quad [\rho(u \wedge v), \rho(x \wedge y)] = \rho([u \wedge v, x \wedge y])$$

We first observe the lemma.

Lemma 2.4. *One has $[\rho(u \wedge v), x] = T_{u \wedge v}(x)$ for every $x \in \mathcal{Cl}(V, q)$.*

PROOF. First, assume that $x \in V$. We use the fundamental identity $uv + vu = -2q(u, v)1$ a few times to see:

$$\begin{aligned} [\rho(u \wedge v), x] &= \frac{1}{4}(uvx - vux - xuv + xvu) \\ &= \frac{1}{2}(-vux + xvu) \\ &= \frac{1}{2}(vxu + 2q(u, x)v - vxu - 2q(v, x)u) \\ &= q(u, x)v - q(v, x)u \\ &= T_{u \wedge v}(x). \end{aligned}$$

□

From this lemma we have

$$[\rho(u \wedge v), \rho(x \wedge y)] = T_{u \wedge v}(\rho(x \wedge y)) = \rho(T_{u \wedge v}(x \wedge y)) = \rho([u \wedge v, x \wedge y]).$$

2.3. Note that by proposition 2.1 that for any $v \in V$ the adjoint action Ad_v preserves the subspace $V \subset \text{Cl}(V, q)$. We define $P(V, q)$ to be the subgroup of $\text{Cl}(V, q)^\times$ generated by vectors $v \in V$ with $q(v) \neq 0$. Let $SP(V, q) = P(V, q) \cap \text{Cl}^{\text{even}}(V, q)$. The group $P(V, q), SP(V, q)$ have important subgroups.

Definition 2.5. The *pin group* of (V, q) is the subgroup $\text{Pin}(V, q) \subset P(V, q)$ generated by elements $v \in V$ with $q(v) = \pm 1$. The *spin group* of (V, q) is

$$(24) \quad \text{Spin}(V, q) = \text{Pin}(V, q) \cap \text{Cl}^{\text{even}}(V, q).$$

Explicit presentation for the pin and spin groups are as follows:

$$\begin{aligned} \text{Pin}(V, q) &= \{v_1 \cdots v_k \in P(V, q) \mid q(v_j) = \pm 1 \ \forall j\} \\ \text{Spin}(V, q) &= \{v_1 \cdots v_k \in \text{Pin}(V, q) \mid k \text{ even}\} \end{aligned}$$

From proposition 2.1, we recognize that $\text{Ad}_v = -R_v$ where R_v is the reflection across the hyperplane perpendicular to $v \in V$. Define the *twisted* adjoint action

$$\widetilde{\text{Ad}}: \text{Cl}(V, q)^\times \rightarrow \text{GL} \text{Cl}(V, q)$$

by the formula

$$(25) \quad \widetilde{\text{Ad}}_\varphi(x) = \alpha(\varphi) x \alpha^{-1},$$

where α is defined in 1.4. Note that $\widetilde{\text{Ad}}_a$ is *not* an algebra automorphism, but it is still a linear automorphism. Notice that for $v \in V$ one has $\widetilde{\text{Ad}}_v = R_v$ as desired.

Proposition 2.6. Define

$$\widetilde{P}(V, q) \stackrel{\text{def}}{=} \{\varphi \in \text{Cl}(V, q) \mid \text{Im } \widetilde{\text{Ad}}_\varphi = V\}.$$

Then the kernel of the homomorphism

$$\widetilde{\text{Ad}}: \widetilde{P}(V, q) \rightarrow \text{GL}(V)$$

is the group $\mathbf{k}^\times$ of nonzero multiples of $1 \in \text{Cl}(V, q)$.

Moreover, $\widetilde{\text{Ad}}$ factors through the group $O(V, q) \subset \text{GL}(V)$.

The next section is dedicated to the proof of this proposition.

2.4. For $a \in \text{Cl}(V, q)$ write $\varphi = \varphi_+ + \varphi_-$ where $\varphi_{\pm} \in \text{Cl}^{\text{ev/odd}}(V, q)$. Then, the condition that $\varphi \in \ker \widetilde{\text{Ad}}$ becomes the pair of equations

$$(26) \quad v\varphi_+ = \varphi_+v, \quad v\varphi_- = -\varphi_-v.$$

Let $\{e_i\}$ be a basis for V such that $q(e_i) \neq 0$ for all i and $\langle e_i, e_j \rangle = 0$ for all $i \neq j$. Using the fundamental Clifford relation, we see that $\varphi_+ \in \text{Cl}^{\text{ev}}(V, q)$ can be expressed in the form $a_0 + e_1a_1$ where a_0, a_1 are polynomial expressions in the basis elements $e_2, \dots, e_n$. Since $a_0 + e_1a_1$ is even we conclude that a_0 is even and a_1 is odd. Applying the relation (26) to $v = e_1$ we see that

$$\begin{aligned} e_1a_0 + e_1^2a_1 &= a_0e_1 + e_1a_1e_1 \\ &= e_1a_0 - e_1^2a_1. \end{aligned}$$

Thus $e_1^2a_1 = 0$ and so $a_1 = 0$. This implies that φ_+ is a polynomial expression in $\{e_2, \dots, e_n\}$. Proceeding iteratively we see that φ is a polynomial expression in *none* of the basis elements, therefore $\varphi_+ \in \mathbf{k} \subset \text{Cl}^{\text{even}}(V, q)$. Similarly, one sees that φ_- is an expression in none of the basis elements. But, since φ_- is odd this implies that $\varphi_- = 0$. Since $\varphi \neq 0$ we conclude that $\varphi \in \mathbf{k}^\times$. We have shown $\ker \widetilde{\text{Ad}} = \mathbf{k}^\times \subset \widetilde{P}(V, q)$.

To complete the proof we introduce the norm mapping. Let N be the linear endomorphism on the Clifford algebra defined by $N(\varphi) = \varphi \cdot \alpha(\varphi^t)$. Note that

$$\begin{aligned} N(\varphi\psi) &= \varphi\psi\alpha(\psi^t\varphi^t) \\ &= \varphi\psi\alpha(\psi^t)\alpha(\varphi^t) \\ &= \varphi N(\psi)\alpha(\varphi^t). \end{aligned}$$

So, we cannot yet conclude that N is compatible with the algebra structure.

Observe for $v \in V$ that $N(v) = -v^2 = q(v)$. Suppose $\varphi \in \widetilde{P}(V, q)$, so that

$$(27) \quad \alpha(\varphi)v\varphi^{-1} \in V$$

for all $v \in V$. Applying the transpose to this element, which is the identity of course, leads to

$$(28) \quad (\varphi^t)^{-1}v\alpha(\varphi^t) = \alpha(\varphi)v\varphi^{-1}.$$

Rearranging, we see that

$$\begin{aligned} v &= \varphi^t\alpha(\varphi)v\varphi^{-1}(\alpha(\varphi^t))^{-1} = \alpha(\alpha(\varphi^t)\varphi)v(\alpha(\varphi^t)\varphi)^{-1} \\ &= \widetilde{\text{Ad}}_{\alpha(\varphi^t)\varphi}(v). \end{aligned}$$

Hence $\alpha(\varphi^t)\varphi \in \ker \widetilde{\text{Ad}} = \mathbf{k}^\times$. We conclude that N factors through the group of units $\mathbf{k}^\times \subset \text{Cl}(V, q)^\times$:

$$(29) \quad N: \widetilde{P}(V, q) \rightarrow \mathbf{k}^\times.$$

This finally allows us to see that N is compatible with the algebra structure. Indeed, since $\mathbf{k}^\times$ is in the center of $\text{Cl}(V, q)$ we have that $N(\varphi\psi) = \varphi N(\psi)\alpha(\varphi^t) = N(\varphi)N(\psi)$.

Notice that $N(\alpha\varphi) = \alpha(\varphi)\varphi^t = N(\varphi)$ for all $\varphi \in \widetilde{P}(V, q)$. Then

$$\begin{aligned} q(\widetilde{\text{Ad}}_\varphi(v)) &= N(\widetilde{\text{Ad}}_\varphi(v)) = N(\alpha(\varphi)v\varphi^{-1}) \\ &= N(\alpha\varphi)N(v)N(\varphi)^{-1} \\ &= q(v). \end{aligned}$$

We conclude that $\widetilde{\text{Ad}}_\varphi$ preserves q for each $\varphi \in \widetilde{P}(V, q)$ so it is an orthogonal transformation.

2.5. By restricting along $P(V, q) \subset \widetilde{P}(V, q)$, proposition 2.6 prescribes a group homomorphism

$$(30) \quad \widetilde{\text{Ad}}: P(V, q) \rightarrow O(V, q).$$

We study the further restriction to $\text{Pin}(V, q)$. The Cartan-Dieudonné theorem implies that the restriction of this homomorphism to $\text{Pin}(V, q)$ is surjective. Similarly, the restriction of $\widetilde{\text{Ad}}$ to $\text{Spin}(V, q)$ defines a surjective homomorphism

$$(31) \quad \widetilde{\text{Ad}}: \text{Spin}(V, q) \rightarrow SO(V, q).$$

Proposition 2.7. *Suppose $\mathbf{k} = \mathbf{R}$. The following sequences are exact*

$$(32) \quad 1 \rightarrow \mathbf{Z}/2 \rightarrow \text{Pin}(V, q) \rightarrow O(V, q) \rightarrow 1$$

and

$$(33) \quad 1 \rightarrow \mathbf{Z}/2 \rightarrow \text{Spin}(V, q) \rightarrow SO(V, q) \rightarrow 1.$$

PROOF. Cartan and Dieudonné did the hard part of surjectivity. From proposition 2.6 if $a \in P(V, q)$ and $\widetilde{\text{Ad}}_a = \mathbb{1}$ then $a = a_0\mathbb{1}$, $a_0 \in \mathbf{R}^\times$. If a is in $\text{Pin}(V, q)$ then we also have $q(a) = \pm 1$, so $a_0 = \pm 1$. The same argument holds for $\text{Spin}(V, q)$. $\square$

2.6. Let's focus on the special case $V = \mathbf{R}^n$ with $q = \sum x_i^2$ the standard positive definite inner product. We let $\mathcal{C}\ell_n \stackrel{\text{def}}{=} \mathcal{C}\ell(\mathbf{R}^n, \sum x_i^2)$, $SO(n) = SO(\mathbf{R}^n, \sum x_i^2)$, and $Spin(n) = Spin(\mathbf{R}^n, \sum x_i^2)$. By the above, for $n \geq 3$ there is a short exact sequence of Lie groups

$$(34) \quad 1 \rightarrow \mathbf{Z}/2 \rightarrow Spin(n) \rightarrow SO(n) \rightarrow 1.$$

THEOREM 2.8. *For $n \geq 3$, the exact sequence (34) represents the universal double cover of $SO(n)$.*

Recall that the universal double cover of a connected topological group G is a covering space

$$(35) \quad 1 \rightarrow \pi_1(H) \rightarrow G \rightarrow H \rightarrow 1$$

where G is the group of equivalence classes of homotopy classes of paths in H with pointwise multiplication. For Lie groups, the universal cover is even more constrained. A basic fact from Lie theory is that any Lie algebra $\mathfrak{g}$ is the Lie algebra of a simply connected Lie group G . Thus, the universal cover of a connected Lie group H is a simply connected Lie group G together with a homomorphism $\rho: G \rightarrow H$ which induces an isomorphism at the level of Lie algebras. Already from the short exact sequence of Lie groups in proposition 2.7 we see that $\widetilde{\text{Ad}}$ induces an isomorphism at the level of Lie algebras.

To prove the theorem we proceed with the following steps.

- (1) First, we will show that $\pi_1(SO(n)) = \pi_1(SO(n+1))$ for $n \geq 3$.
- (2) Next, we will show that $\pi_1(SO(3)) = \mathbf{Z}/2$ by showing $Spin(3) = SU(2)$.
- (3) Finally, we will argue that $\pi_1(Spin(n)) = 0$ for $n \geq 3$, thus completing the proof.

For (1) we use the long exact sequence in homotopy groups associated to the fibration

$$(36) \quad SO(n) \hookrightarrow SO(n+1) \rightarrow S^n$$

induced from embedding $SO(n)$ block diagonally into $SO(n+1)$.

For (2) we first construct an explicit isomorphism between the Lie algebra $\mathfrak{su}(2)$ of $SU(2)$ and $\mathbf{R}^3$. Explicitly $\mathfrak{su}(2)$ is the Lie algebra of complex 2×2 matrices of the form

$$(37) \quad \begin{pmatrix} a & -\bar{b} \\ b & -a \end{pmatrix}.$$

In terms of matrices, the standard positive definite inner product on $\mathbf{R}^3$ is

$$(38) \quad \frac{1}{2} \text{Tr}(XY^\dagger).$$

Now, define the homomorphism

$$(39) \quad \rho: SU(2) \rightarrow GL(\mathfrak{su}(2)) \cong GL(\mathbf{R}^3)$$

by sending the matrix A to $X \mapsto AXA^{-1}$. This matrix factors through $SO(3)$. It is surjective and its kernel is $\mathbf{Z}/2$. Since $SU(2)$ is diffeomorphic to S^3 it follows that $Spin(3) \cong SU(2)$ and $\pi_1(SO(3)) = \mathbf{Z}/2$. With this identification $\rho = \widetilde{\text{Ad}}$.

Finally, to see (3) we recall that covering space theory implies that $\pi_1(Spin(n))$ is a finite index subgroup of $\mathbf{Z}/2$. So, for $n \geq 3$ one sees that $\pi_1(Spin(n))$ is trivial.

3. Low-dimensional examples

We will present some basic low-dimensional examples of real Clifford algebras and spin groups. We let $Cl_{r,s}$ denote the Clifford algebra of the vector space $\mathbf{R}^{r+s}$ associated to the quadratic form of signature (r,s) . (In terms of our previous notation $Cl_{n,0} = Cl_n$.)

3.1. The Clifford algebra Cl_1 is generated by elements $1, e$ with the relation $e^2 = -1$. Thus $Cl_1 \cong \mathbf{C}$ as real associative algebras. Under this identification, $Cl_1^{\text{ev}} = \mathbf{R}$ and $Cl_1^{\text{odd}} = i\mathbf{R}$. The transpose operation is the identity. The map α is complex conjugation $\alpha(z) = \bar{z}$. The group of units is the nonzero complex numbers under multiplication $Cl_1^\times = \mathbf{C}^\times$. The norm map is $N(z) = z\bar{z}$.

We know from the exact sequences from proposition 2.7 that

$$(40) \quad Pin(1) \simeq \mathbf{Z}/4, \quad Spin(1) \simeq \mathbf{Z}/2.$$

Let's see this explicitly. Per the isomorphisms of the previous section, we can identify $Pin(1)$ with the group of elements $z = a + ib \in \mathbf{C}^\times$ such that $a = \pm 1, b = 0$ or $a = 0, b = \pm 1$. Thus $Pin(1) = \{1, -1, i, -i\} = \mathbf{Z}/4$ and $Spin(1) = \{1, -1\} = \mathbf{Z}/2$.

3.2. Next we look at Cl_2 . Let $\{e_1, e_2\}$ be an orthonormal basis for $V = \mathbf{R}^2$. Then Cl_2 is spanned by the basis $\{1, e_1, e_2, e_1e_2\}$ subject to the relations

$$(41) \quad e_1e_2 = -e_2e_1, \quad e_1^2 = e_2^2 = -1, \quad (e_1e_2)^2 = -1.$$

Define the real linear map

$$(42) \quad \Phi: Cl_2 \rightarrow \mathbf{H}$$

by the rules $e_1 \mapsto i, e_2 \mapsto j, e_1 e_2 \mapsto k$. It is immediate to check that this is an isomorphism of real algebras. Thus $C\ell_2$ is isomorphic to the quaternions, which is of course generated over $\mathbf{R}$ by $\{1, i, j, k\}$ satisfying the usual conditions.

In quaternion terms the transpose is

$$(43) \quad 1^t = 1, \quad i^t = i, \quad j^t = j, \quad k^t = -k.$$

The involution α is

$$(44) \quad \alpha(1) = 1, \quad \alpha(i) = -i, \quad \alpha(j) = -j, \quad \alpha(k) = k.$$

In particular, $1, k$ are even and i, j are odd. The norm is

$$(45) \quad N(1) = N(i) = N(j) = N(k) = 1.$$

The group $Pin(2)$ thus consists of elements

$$(46) \quad a1 + bi + cj + dk, \quad a, b, c, d \in \mathbf{R}$$

such that

- Either $b = c = 0$ and $a^2 + d^2 = 1$, or
- $a = d = 0$ and $b^2 + c^2 = 1$.

We conclude that $Pin(2) \simeq U(1) \sqcup U(1)$ and $Spin(2) \simeq U(1)$.

In quaternion notation, the group $Spin(2) \simeq U(1)$ consists of elements $a1 + dk \in \mathbf{H}$ satisfying $a^2 + d^2 = 1$. In terms of a real orthonormal basis of $\mathbf{R}^2$, this group is presented as the elements

$$(47) \quad x = a1 + be_1e_2$$

satisfying $N(x) = a^2 + b^2 = 1$.

3.3. The Clifford algebra $Cl_{0,2}$ is spanned by vectors $1, x, y, xy$ which satisfy

$$(48) \quad x^2 = y^2 = 1, \quad xy = -yx.$$

The correspondence

$$\begin{aligned} 1 &\mapsto \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \\ x &\mapsto \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \\ y &\mapsto \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \end{aligned}$$

sets up an isomorphism of algebras between $Cl_{0,2}$ and the algebra of 2×2 real matrices.

For future reference, if $\mathbf{k}$ is a field, we will denote $\mathbf{k}(n)$ for the algebra of $n \times n$ matrices with coefficients in $\mathbf{k}$.

Thus $Cl_{0,2} \cong \mathbf{R}(2)$.

3.4. The Clifford algebra $Cl_{1,1}$ is spanned by vectors $1, x, y, xy$ which satisfy

$$(49) \quad x^2 = -y^2 = 1, \quad xy = -yx.$$

The correspondence

$$\begin{aligned} 1 &\mapsto \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \\ x &\mapsto \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \\ y &\mapsto \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \end{aligned}$$

sets up an isomorphism of algebras between $Cl_{1,1}$ and the algebra of 2×2 real matrices.

4. Classification of Clifford algebras

The main result of this section is to prove the following periodicity result for Clifford algebras

$$(50) \quad Cl_{n+4} \simeq Cl_n \otimes Cl_4.$$

4.1.

Proposition 4.1. *There are isomorphism*

$$(51) \quad \begin{aligned} \text{Cl}_{n+2,0} &\simeq \text{Cl}_{0,n} \otimes \text{Cl}_{2,0} \\ \text{Cl}_{0,n+2} &\simeq \text{Cl}_{n,0} \otimes \text{Cl}_{0,2} \\ \text{Cl}_{n+1,m+1} &\simeq \text{Cl}_{n,m} \otimes \text{Cl}_{1,1}. \end{aligned}$$

The tensor product is the ordinary (ungraded) tensor product of algebras.

PROOF. Let $\{e_i\}$ denote an orthonormal basis for $\mathbf{R}^{n+2}$ with respect to the standard positive definite form. Let $\{e'_i\}$ be an orthonormal basis for $\mathbf{R}^n$ which we view as generators for the algebra $\text{Cl}_{0,n}$. In particular $e'_i e'_i = 1$ (note the lack of sign). Let $\{e''_1, e''_2\}$ be an orthonormal basis for $\mathbf{R}^2$ which we view as generators for $\text{Cl}_{2,0}$. Define the linear map

$$(52) \quad f: \mathbf{R}^{n+2} \rightarrow \text{Cl}_{0,n} \otimes \text{Cl}_{2,0}$$

by the following rules. If $i = 1, \dots, n$ then define $f(e_i) = e'_i \otimes e''_1 e''_2$. Additionally $f(e_{n+1}) = 1 \otimes e''_1$ and $f(e_{n+2}) = 1 \otimes e''_2$. This map f satisfies $f(v)^2 = -\|v\|^2$ where $\| - \|$ is the ordinary norm on $\mathbf{R}^{n+2}$. Thus f extends to a homomorphism $\tilde{f}: \text{Cl}_{n+2,0} \rightarrow \text{Cl}_{0,n} \otimes \text{Cl}_{2,0}$. Since f hits all generators it follows that $\tilde{f}$ is surjective. By counting dimensions we see its an isomorphism. $\square$

4.2. Using the proposition of the previous section we can prove the stated periodicity result.

THEOREM 4.2. *There is an isomorphism of real algebras*

$$(53) \quad \text{Cl}_{n+4} \simeq \text{Cl}_n \otimes \text{Cl}_4$$

for every n . The tensor product is the ordinary (ungraded) tensor product.

We note that there are similar periodicity results for the non-definite signature Clifford algebras.

PROOF. From the proposition of the previous subsection

$$(54) \quad \text{Cl}_{4,0} \simeq \text{Cl}_{0,2} \otimes \text{Cl}_{2,0} \simeq \mathbf{H} \otimes \mathbf{R}(2) \simeq \mathbf{H}(2) \simeq \text{Cl}_{2,0} \otimes \text{Cl}_{0,2} \simeq \text{Cl}_{0,4}$$

Thus using the proposition again we have

$$(55) \quad \begin{aligned} \text{Cl}_n \otimes \text{Cl}_{4,0} &\simeq \text{Cl}_n \otimes \text{Cl}_{0,2} \otimes \text{Cl}_{2,0} \\ &\simeq \text{Cl}_{0,n+2} \otimes \text{Cl}_{2,0} \\ &\simeq \text{Cl}_{n+4,0} \end{aligned}$$

as desired. □

4.3. From the theorem we see that to describe explicitly $\mathcal{Cl}_n$ for any n it suffices to know $\mathcal{Cl}_0 = \mathbf{k}, \mathcal{Cl}_1 = \mathbf{C}, \mathcal{Cl}_2 = \mathbf{H}, \mathcal{Cl}_3$ and $\mathcal{Cl}_4$. We already computed $\mathcal{Cl}_4 = \mathbf{H}(2)$ in the proof of the theorem. Finally $\mathcal{Cl}_3 = \mathcal{Cl}_{0,1} \otimes \mathcal{Cl}_{2,0} = \mathbf{H} \oplus \mathbf{H}$.

4.4. Similarly there is the complex version of the above isomorphism.

THEOREM 4.3. *There is an isomorphism of complex algebras*

$$(56) \quad \mathbf{Cl}_{n+2} \simeq \mathbf{Cl}_n \otimes \mathbf{Cl}_2.$$

The tensor product is also ungraded.

References

- [LM89] H. B. Lawson Jr. and M.-L. Michelsohn. *Spin geometry*. Vol. 38. Princeton Mathematical Series. Princeton University Press, Princeton, NJ, 1989, pp. xii+427.